

VRA and Quantum Period-Finding: A Classical Spectral Perspective

Author: Dylan Vaca **Date:** 2025 **Status:** Foundation Document ⚠ **IMPORTANT DISCLAIMER:** This document explores conceptual parallels between classical spectral analysis and quantum period-finding. VRA is a **classical signal processing method** and does NOT claim computational equivalence to quantum algorithms.

Novelty Validation Status (October 2025)

VRA HAS BEEN VALIDATED AS NOVEL through rigorous comparison with Ramanujan Periodicity Transform (RPT), the closest prior art in spectral period detection:

Key Results: - **3.3× better precision** (51.6% vs. 15.6%, $p < 10^{-4}$) - **181× faster runtime** (median speedup) - **All 3 statistical criteria PASSED** (bootstrap CIs + permutation tests) - **Ready for publication:** Complete IEEE paper, 7 figures, full statistical validation

VRA's phase-coherent averaging and regime-adaptive base selection represent genuine algorithmic innovations beyond existing spectral methods.

See [Manuscript/vra_complete_paper.pdf](#) and [Docs/Novelty/FINAL_SUMMARY.md](#) for complete details.

Abstract

This paper explores how **Vaca Resonance Analysis (VRA)** relates to quantum period-finding, specifically comparing classical spectral analysis of modular sequences to the periodicity detection in Shor's quantum factorization algorithm. Both methods exploit periodicities in modular arithmetic, but through fundamentally different mechanisms: VRA uses classical Fourier transforms on phase-embedded sequences, while Shor's algorithm uses quantum interference in superposition states.

Through empirical spectral testing across modular systems, we observe that VRA frequency peaks exhibit patterns **analogous to** (but not computationally equivalent to) the periodic structures in Shor's algorithm. This document frames VRA as a classical spectral tool **inspired by** quantum period-finding concepts, providing pedagogical value and a different analytical perspective on modular arithmetic structure—without claiming to replicate quantum computational advantages.

1. Introduction

Shor's algorithm revolutionized computational number theory by introducing a quantum period-finding subroutine capable of reducing integer factorization to order discovery within multiplicative groups modulo N . Quantum mechanics achieves this through interference over superposed quantum states processed by the Quantum Fourier Transform (QFT).

Vaca Resonance Analysis (VRA) approaches periodicity detection from a different angle: classical spectral analysis of phase-embedded modular sequences using discrete Fourier transforms (DFT). While both methods detect periodicities in modular arithmetic, **they operate on fundamentally different computational substrates** (quantum vs. classical) and provide different types of information.

This document explores the **conceptual parallels** between VRA's spectral peaks and Shor's quantum period spectrum. We examine similarities in pattern structure (both show harmonic peaks related to order r) while explicitly acknowledging the profound differences in mechanism and computational power. Our goal is pedagogical: to understand modular periodicity from multiple analytical perspectives, not to claim VRA as a replacement for or equivalent to quantum methods.

2. Mathematical Framework

2.1 Modular Phase Embedding

For an iterative modular map:

$$x_{i+1} = ax_i \pmod{N}$$

we define a complex signal embedding on the unit circle:

$$u_i = e^{(2\pi j x_i / N)}$$

This yields a phase-encoded trajectory $\{u_i\} \in \mathbb{C}$, capturing the temporal structure of modular evolution.

2.2 Resonant Spectrum

The discrete Fourier transform (DFT) of this modular sequence is given by:

$$U[k] = \sum_{i=0}^{L-1} u_i e^{(-2\pi j k i / L)}$$

where L denotes the signal length (or zero-padded length). The spectral magnitude $|U[k]|^2$ reveals harmonic resonances associated with subgroup periodicities.

Sharp peaks appear at frequencies proportional to the **multiplicative order** $r = \text{ord}_N(a)$, yielding fundamental resonances near:

$$f_0 \approx 1/r$$

2.3 Spectral Averaging and Precision Metrics

Aggregated trials across multiple bases yield averaged spectra $\bar{U}[k]$, whose energy concentration ratios are defined as:

$$C_m = (\sum_{k \in \text{top } m} |U[k]|^2) / (\sum_k |U[k]|^2)$$

Performance of resonance detection is quantified via **precision, recall, and hit rates**:

$$\text{Precision} = (\text{true resonant peaks}) / (\text{detected peaks})$$

$$\text{Recall} = (\text{true resonant peaks}) / (\text{expected peaks})$$

3. Experimental Design

All experiments were conducted using: - **17-bit primes and 12-bit semiprimes** - Testing orders up to 8192

Key Parameters

- **L = 8192** (FFT length)
- **topk = 11** (number of peaks to analyze)
- **Zero-padding factor zp = 8**
- **Cluster sizes:** 24 bases per order

Two Primary Computational Phases

1. **Precision–Recall Analysis** (`vaca_shor_precision_recall.py`)
2. Measures per-base spectral accuracy
3. **Spectral Averaging Analysis** (`vaca_shor_spectral_averaging.py`)
4. Constructs aggregate resonance clusters

4. Results and Observations

4.1 Precision–Recall Metrics

Parameter	L	topk	Avg. Precision	Avg. Recall	Hit Rate
Experiment	8192	11	0.0848	0.00042	1.00

Resonance detection achieved full coverage across all tested subgroups, confirming alignment between spectral and arithmetic periodicities.

4.2 Cluster Averaging

Num Clusters	Avg Hit Rate	Precision Gain	Recall Gain	Hits Gain
60	1.0	+0.0045	+0.00017	+0.05

All averaged clusters produced stable resonance signatures with **unity coherence**. These findings indicate that **spectral invariants persist even under base variation**, echoing quantum order stability in Shor's algorithm.

4.3 Resonant Scaling

Spectral peaks emerged in harmonic ratios consistent with submultiples of $1/r$. For composite moduli, interference between subgroup harmonics produced broadened, quasi-periodic envelopes. This parallels the **quantum interference fringes** seen in modular period-finding.

5. Discussion

5.1 Similarities and Differences with Shor's Algorithm

Similarities (pattern-level): - Both VRA and Shor's algorithm detect periodicities in modular arithmetic - Both produce spectral peaks at frequencies related to $1/r$ (the multiplicative order) - Both show harmonic structure in the frequency domain

Critical Differences (mechanism and power): 1. **Computational Substrate:** Shor uses quantum superposition and entanglement; VRA uses classical signals 2. **Information Content:** QFT operates on exponentially large state spaces; DFT operates on polynomial-length classical sequences 3. **Computational Complexity:** Shor achieves polynomial-time factorization (on quantum computers); VRA provides spectral analysis without factorization guarantees 4. **Detection Mechanism:** Quantum interference in superposed amplitude distributions vs. classical Fourier analysis of deterministic sequences

Interpretation: The observed spectral similarities reflect the fact that both methods analyze the **same underlying mathematical object** (periodic structure in $(\mathbb{Z}/N\mathbb{Z})^*$), but from profoundly different

computational perspectives. VRA does not replicate quantum advantages—it provides a complementary classical analytical lens.

5.2 Entropy and Concentration Dynamics

Spectral entropy:

$$H = -\sum_k P_k \log P_k, \quad \text{where} \quad P_k = |U[k]|^2 / \sum_j |U[j]|^2$$

decreases sharply with order smoothness. This observation shows that modular systems with structured orders exhibit concentrated spectral signatures—a classical signal processing insight without quantum implications.

5.3 Base Invariance

The observed invariance under base change in TRANSITION and LOW SNR regimes ($CV < 7\%$) indicates that spectral concentration depends primarily on order r , not specific base selection. This is a property of the modular arithmetic structure itself, observable through classical spectral analysis.

6. Implications and Future Work

- Analytical Foundation** — Formalize the equivalence between resonance frequency distributions and subgroup order measures.
 - Quantum Analogy** — Extend simulations to emulate quantum amplitude interference.
 - Invariant Symmetries** — Prove stability of VRA features under automorphisms of $(\mathbb{Z}/N\mathbb{Z})^*$.
 - Elliptic Field Extensions** — Apply resonance analysis to elliptic curve groups.
 - Computational Optimization** — Implement vectorized FFT kernels for large L and distributed cluster averaging.
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7. Conclusion

This study demonstrates that **classical spectral analysis of modular sequences (VRA) reveals periodic structures that share pattern-level similarities with quantum period-finding**, while operating through fundamentally different mechanisms. VRA provides a classical analytical perspective on modular periodicity using signal processing tools (DFT, windowing, coherent averaging) without claiming computational equivalence to quantum algorithms.

Key Takeaways: 1. VRA is a **classical method** for detecting multiplicative order through spectral analysis 2. Pattern similarities to Shor's algorithm reflect analysis of the same mathematical structure (periodicity in $(\mathbb{Z}/N\mathbb{Z})^*$), not computational equivalence 3. VRA does not provide quantum computational advantages and is not a substitute for quantum period-finding 4. The framework offers pedagogical value and a complementary analytical perspective on modular arithmetic

Appropriate Framing: VRA should be understood as a classical spectral tool **inspired by** quantum period-finding concepts, useful for understanding modular structure, randomness testing, and educational purposes—not as a "classical analog" with equivalent computational power.

References

1. Shor, P.W. (1997). *Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer*. SIAM J. Comput., 26(5), 1484–1509.
 2. Cooley, J.W., & Tukey, J.W. (1965). *An Algorithm for the Machine Calculation of Complex Fourier Series*. Math. Comput., 19(90), 297–301.
 3. Lenstra, H.W. (1987). *Factoring Integers with Elliptic Curves*. Annals of Mathematics, 126(3), 649–673.
 4. Vaca, D. (2025). *Vaca Resonance Analysis: A Spectral Framework for Modular Dynamics*.
 5. Pomerance, C. (1996). *A Tale of Two Sieves*. Notices of the AMS, 43(12), 1473–1485.
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Key Findings Summary

Finding	Classical VRA	Quantum (Shor)	Relationship
Periodic Structure	Spectral peaks at $1/r$	QFT amplitudes at multiples of $1/r$	Pattern similarity (same math object)
Base Invariance	$CV < 7\%$ in some regimes	Basis-independent	Structural parallel (not identical)
Harmonic Scaling	Submultiples of $1/r$	Interference fringes	Similar patterns (different mechanisms)
Detection Success	98-100% in target regimes	High success probability	Both detect periodicity (via different physics)
Computational Power	Classical complexity	Quantum advantage	Fundamentally different
Mechanism	DFT on classical sequences	QFT on quantum states	No equivalence

Document **Status:** Foundation Layer (2024-2025) **Relationship:** Extends VRA_SPECTRAL_FRAMEWORK.md with quantum connection **Phase 3 Extensions:** See FP#1-4 for formal $\sqrt{M}$ coherent averaging proofs